$= 6.2 \times 10^{-20}$ J (accept 1 sig.fig.) .... A1 [2]

4	**(a)(i)**	$\omega = 2\pi f$	C1	
		$= 2\pi \times 1400$		
		$= 8800$ rad s^{-1}	A1	**[2]**
	(ii)	$a_0 = (-)\omega^2 x_0$	C1	
		$= (8800)^2 \times 0.080 \times 10^{-3}$		
		$= 6200$ m s^{-2}	A1	**[2]**
	(b)	straight line through origin with negative gradient	M1	
		end points of line correctly labelled	A1	**[2]**
	(c)(i)	zero displacement	B1	**[1]**
	(ii)	$v = \omega x_0$	C1	
		$= 8800 \times 0.080 \times 10^{-3}$		
		$= 0.70$ m s^{-1}	A1	**[2]**